CCNA Exam Prep: Back to Networking Basics with Hank Preston and Patrick Gargano -- Season 4

OSPFv3 Unlocked: Routing the IPv6 Way

Related CCNA v2.0 exam topics:

- 1.4: Troubleshoot IPv6 address configuration, assignment, and prefix sizing
- 3.3: Configure single area OSPFv2 for IPv4 and OSPFv3 for IPv6
  - 3.3.a Neighbor adjacencies (excluding authentication)
  - 3.3.b Point-to-point
  - 3.3.c Broadcast (DR/BDR selection)
  - 3.3.d Router ID
- 4.7.d: Configure Layer 2 security features: RA guard

In this lab, you will explore how to:

- Configure and verify single-area OSPFv3
- Tune OSPFv3 (OSPF cost, DR/BDR election, passive interface)
- Observe legitimate IPv6 SLAAC and Router Advertisements (RA)
- Observe a rogue IPv6 RA injected into an access VLAN
- Configure and verify IPv6 RA Guard to block the rogue RA.

Setup and Scenario

You administer a small enterprise running IPv6-only routing. R1, R2, and R3 form the backbone; SW1 is the default gateway for the user VLAN (VLAN 20); R5 is a branch router.

During a cabling change, R5's second interface is mistakenly patched into an access port on SW1 in VLAN 20. Because R5 has `ipv6 unicast-routing` enabled, it immediately begins sending RAs advertising an incorrect prefix and offering itself as a default gateway. A SLAAC host in VLAN 20 (simulated by E0/2 on R2) begins autoconfiguring the incorrect prefix and loses connectivity to the network.

Your job: build and verify the OSPFv3 network, reproduce the rogue-RA condition, then harden SW1 with RA Guard so only it may advertise RAs into VLAN 20. In summary:

- SW1 SVI 20 is the authorized VLAN 20 gateway
- R5 Ethernet0/1 is deliberately mis-patched into VLAN 20 as a rogue RA source
- R2 Ethernet0/2 acts as the SLAAC victim host
- IPv6 RA Guard on SW1 mitigates the attack

Roles

The network is designed with the following roles:

- R1, R3: backbone routers
- R2: backbone router; Et0/2 also acts as a VLAN 20 SLAAC HOST
- R5: branch router; Et0/1 is MIS-PATCHED into VLAN 20 (ROGUE)
- SW1: Layer 2 access switch + multilayer gateway (RID 4.4.4.4, SVI 20 = authorized gateway, SVI 100 = uplink to R1)

Addressing

Device	Interface	IPv6 Address	Link-local
R1	Lo0	2001:db8:acad:ffff::1/128	-
R1	Et0/0 -> R2	2001:db8:acad:12::1/64	fe80::1
R1	Et0/1 -> R3	2001:db8:acad:13::1/64	fe80::1
R1	Et0/2 -> SW1	2001:db8:acad:14::1/64	fe80::1
R2	Lo0	2001:db8:acad:ffff::2/128	-
R2	Et0/0 -> R1	2001:db8:acad:12::2/64	fe80::2
R2	Et0/1 -> R3	2001:db8:acad:23::2/64	fe80::2
R2	Et0/2 -> SW1	autoconfig (SLAAC host)	fe80::2
R3	Lo0	2001:db8:acad:ffff::3/128	-
R3	Et0/0 -> R2	2001:db8:acad:23::3/64	fe80::3
R3	Et0/1 -> R1	2001:db8:acad:13::3/64	fe80::3
R3	Et0/2 -> R5	2001:db8:acad:35::3/64	fe80::3
R5	Lo0	2001:db8:acad:ffff::5/128	-
R5	Et0/0 -> R3	2001:db8:acad:35::5/64	fe80::5
R5	Et0/1 -> SW1	2001:db8:bad:20::1/64 ROGUE	fe80::5
SW1	Lo0	2001:db8:acad:ffff::4/128	-
SW1	Vlan100	2001:db8:acad:14::4/64	fe80::4
SW1	Vlan20	2001:db8:acad:20::1/64	fe80::4

SW1 Port Map

- Et0/0 = VLAN 100 uplink to R1 (trusted, device-role **router**)
- Et0/1 = VLAN 20 access, R2 host (device-role **host**)
- Et0/2 = VLAN 20 access, R5 ROGUE (device-role **host** -> RAs dropped)
- Et0/3 = unused / shut

Boot state

- R2 Ethernet 0/2 is administratively DOWN at boot. Bring it up only in Part 3 when told to do so. This interface simulates a legitimate user in VLAN 20.
- R5 Ethernet0/1 is administratively DOWN at boot. Bring it up only in Part 4 when told to do so. This interface simulates a rogue source of RAs in VLAN 20.
- All IPv6 addresses (global and link-local) are preconfigured on R1, R2, R3, SW1, and R5.

*Be sure to **START** the lab before continuing*

> Note: The credentials for all devices are **cisco / cisco** (enable secret: **cisco**)

Part 1: Configure and verify single-area OSPFv3

In this part of the lab, you will enable OSPFv3 and manually set the OSPF router ID (RID). OSPFv3 has no automatic router ID when there are no IPv4 addresses, so the RID must be configured manually.

Step 1

By default, IPv6 unicast routing is not enabled on Cisco routers and switches. Start by enabling it on R1, R2, R3, SW1, and R5.

The output below shows the command configured on R1. Repeat the same step on R2, R3, SW1, and R5.

```
R1# configure terminal
R1(config)# ipv6 unicast-routing
```

Step 2

Enable OSPFv3 with process ID 1 and manually set the router ID. Use the following router ID values:

- R1: 1.1.1.1
- R2: 2.2.2.2
- R3: 3.3.3.3
- SW1: 4.4.4.4
- R5: 5.5.5.5

Repeat the following commands on R2, R3, SW1, and R5 using the appropriate router ID value:

```
R1(config)# ipv6 router ospf 1
R1(config-router)# router-id 1.1.1.1
R1(config-router)# exit
```

Step 3

Enable OSPFv3 on R1, R2, R3, SW1, and R5 for the following interfaces:

Device	Interface
R1	Lo0
R1	Et0/0 -> R2
R1	Et0/1 -> R3
R1	Et0/2 -> SW1
R2	Lo0
R2	Et0/0 -> R1
R2	Et0/1 -> R3
R3	Lo0
R3	Et0/0 -> R2
R3	Et0/1 -> R1
R3	S1/0 -> R5
R5	Lo0
R5	S1/0 -> R3
SW1	Lo0
SW1	Vlan100
SW1	Vlan20

Use the OSPFv3 interface-level command `ipv6 ospf 1 area 0` to include the interfaces in area 0 of the OSPFv3 process ID 1.

> Note: **Do not** enable OSPFv3 on the **R5 Et0/1** interface. That interface represents the mis-patched cable and will be used later in Part 4.

> Note: **Do not** enable OSPFv3 on the **R2 Et0/2** interface. That interface represents a SLAAC host in VLAN 20 and will be used later in Part 3.

The following CLI code illustrates the commands required for R1. Repeat these commands on the appropriate interfaces on R2, R3, SW1, and R5 according to the table above.

```
R1(config)# interface Loopback0
R1(config-if)# ipv6 ospf 1 area 0
R1(config-if)# interface Ethernet0/0
R1(config-if)# ipv6 ospf 1 area 0
R1(config-if)# interface Ethernet0/1
R1(config-if)# ipv6 ospf 1 area 0
R1(config-if)# interface Ethernet0/2
R1(config-if)# ipv6 ospf 1 area 0
```

Step 4

Once OSPFv3 is configured on all network devices, verify that the correct number of neighbor adjacencies have been established on R1 and R3.

```
R1# show ipv6 ospf neighbor

            OSPFv3 Router with ID (1.1.1.1) (Process ID 1)
```

```
Neighbor ID     Pri   State           Dead Time   Interface ID    Interface
4.4.4.4           1   FULL/DR         00:00:34    9               Ethernet0/2
3.3.3.3           1   FULL/DR         00:00:33    2               Ethernet0/1
2.2.2.2           1   FULL/DR         00:00:38    1               Ethernet0/0

R3# show ipv6 ospf neighbor

           OSPFv3 Router with ID (3.3.3.3) (Process ID 1)

Neighbor ID     Pri   State           Dead Time   Interface ID    Interface
5.5.5.5           0   FULL/  -        00:00:31    5               Serial1/0
1.1.1.1           1   FULL/BDR        00:00:34    2               Ethernet0/1
2.2.2.2           1   FULL/BDR        00:00:33    2               Ethernet0/0
```

R1 should report three `FULL` OSPFv3 adjacencies, one each for Ethernet0/0, Ethernet0/1, and Ethernet0/2.

R3 should also report three `FULL` OSPFv3 adjacencies, one each for Serial1/0, Ethernet0/0, and Ethernet0/1.

You will tune the DR/BDR election later in Part 2. Currently the election of the DR and BDR is based on the OSPFv3 router ID, with the higher value being the deciding factor since all Ethernet interfaces have a default OSPFv3 priority value of 1. Your DR/BDR results might vary from the output above depending on the sequence you used to enable OSPFv3 on the routers and switch.

Step 5

Next, verify that the correct interfaces are configured for OSPFv3 on R1, R2, R3, SW1, and R5.

```
R1# show ipv6 ospf interface brief
Interface    PID   Area            Intf ID    Cost  State Nbrs F/C
Lo0          1     0               7          1     LOOP  0/0
Et0/2        1     0               4          10    BDR   1/1
Et0/1        1     0               3          10    BDR   1/1
Et0/0        1     0               2          10    BDR   1/1

R2# show ipv6 ospf interface brief
Interface    PID   Area            Intf ID    Cost  State Nbrs F/C
Lo0          1     0               7          1     LOOP  0/0
Et0/1        1     0               3          10    BDR   1/1
Et0/0        1     0               2          10    DR    1/1

R3# show ipv6 ospf interface brief
Interface    PID   Area            Intf ID    Cost  State Nbrs F/C
Lo0          1     0               11         1     LOOP  0/0
Se1/0        1     0               6          64    P2P   1/1
Et0/1        1     0               3          10    DR    1/1
Et0/0        1     0               2          10    DR    1/1

SW1# show ipv6 ospf interface brief
Interface    PID   Area            Intf ID    Cost  State Nbrs F/C
Lo0          1     0               7          1     LOOP  0/0
Vl100        1     0               9          1     DR    1/1
Vl20         1     0               8          1     DR    0/0

R5# show ipv6 ospf interface brief
Interface    PID   Area            Intf ID    Cost  State Nbrs F/C
Lo0          1     0               11         1     LOOP  0/0
Se1/0        1     0               6          64    P2P   1/1
```

All configured interfaces are assigned to area 0. Notice that the **State** column reports the following values:

- Ethernet interfaces are reporting either `DR` or `BDR`
- Serial interfaces are reporting `P2P`
- Loopback interfaces are reporting `LOOP`

By default, OSPFv2 and OSPFv3 treat Ethernet interfaces as a broadcast network. This network type requires the election of a DR and BDR. Serial interfaces are treated as point-to-point links with only two devices sharing the link. No DR or BDR are required on point-to-point links. Loopback interfaces are treated as a stub network with only one host (itself).

Step 5

Verify the IPv6 OSPF routes on R1, R2, R3, SW1, and R5, and test reachability.

Use the `show ipv6 route ospf` command to filter the output to only show OSPFv3 routes.

The output below shows the OSPFv3 routes on R1.

```
R1# show ipv6 route ospf
IPv6 Routing Table - default - 15 entries
Codes: C - Connected, L - Local, S - Static, U - Per-user Static route
       B - BGP, R - RIP, H - NHRP, HG - NHRP registered
       Hg - NHRP registration summary, HE - NHRP External, I1 - ISIS L1
       I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary, D - EIGRP
       EX - EIGRP external, ND - ND Default, NDp - ND Prefix, DCE - Destination
       NDr - Redirect, RL - RPL, O - OSPF Intra, OI - OSPF Inter
       OE1 - OSPF ext 1, OE2 - OSPF ext 2, ON1 - OSPF NSSA ext 1
       ON2 - OSPF NSSA ext 2, la - LISP alt, lr - LISP site-registrations
       ld - LISP dyn-eid, lA - LISP away, le - LISP extranet-policy
       lp - LISP publications, ls - LISP destinations-summary, a - Application
       m - OMP
O   2001:DB8:ACAD:20::/64 [110/11]
     via FE80::4, Ethernet0/2
O   2001:DB8:ACAD:23::/64 [110/20]
     via FE80::3, Ethernet0/1
```

```
     via FE80::2, Ethernet0/0
O   2001:DB8:ACAD:35::/64 [110/74]
     via FE80::3, Ethernet0/1
O   2001:DB8:ACAD:FFFF::2/128 [110/10]
     via FE80::2, Ethernet0/0
O   2001:DB8:ACAD:FFFF::3/128 [110/10]
     via FE80::3, Ethernet0/1
O   2001:DB8:ACAD:FFFF::4/128 [110/10]
     via FE80::4, Ethernet0/2
O   2001:DB8:ACAD:FFFF::5/128 [110/74]
     via FE80::3, Ethernet0/1
```

Note the use of the manually configured link-local addresses as the next-hop for all OSPFv3 entries. All entries fall under the *OSPF Intra-area* category since all interfaces are part of the same area, area 0.

OSPF Cost

Recall that every router runs OSPF's Dijkstra Shortest Path First algorithm against the link-state database to build a loop-free tree rooted at itself. OSPF uses a metric called **cost** to calculate the shortest path to a destination.

Three properties define the OSPF cost:

- Cost is applied per outbound interface. A router assigns cost to traffic leaving an interface, never entering it.
- Cost is cumulative and additive. The metric of a route is the sum of the outbound interface costs along the entire path to the destination network.
- Cost is based on the interface bandwidth. The faster the interface, the lower the cost.

The lowest total cost wins. In our network, the interfaces in use have the following default costs:

- Ethernet: 10
- Serial: 64
- Switch SVI: 1

Looking at R1's routing table, you can easily calculate the costs to reach the prefixes listed.

For example, for R1 to reach the `2001:DB8:ACAD:20::/64` prefix, the total cost is 11 (R1 E0/2 interface cost of 10 plus SW1 VLAN 20 SVI cost of 1).

Notice how there are two next-hop entries for R1 to reach the `2001:DB8:ACAD:23::/64` prefix with both showing a cost of 20. When multiple paths tie on total cost, OSPF installs up to four equal-cost routes in the routing table by default and load-balances across them.

Another cost calculation example would be the cost listed for R1 to reach the `2001:DB8:ACAD:35::/64` prefix. The total cost shown in the routing table is 74 which is based on the cost of R1's E0/1 interface cost of 10 plus the cost of the R3 S1/0 interface of 64.

You will tune the OSPF cost in Part 2 of the lab.

Reachability test

From the R5 router, ping the SW1 VLAN 20 SVI global unicast address and perform a traceroute to the same address.

```
R5# ping 2001:db8:acad:20::1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:DB8:ACAD:20::1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 22/23/28 ms

R5# traceroute 2001:db8:acad:20::1
Type escape sequence to abort.
Tracing the route to 2001:DB8:ACAD:20::1

  1 2001:DB8:ACAD:35::3 21 msec 22 msec 21 msec
  2 2001:DB8:ACAD:13::1 22 msec 22 msec 22 msec
  3 2001:DB8:ACAD:14::4 23 msec 22 msec 23 msec
```

Since there is no OSPFv3 neighbor adjacency between R5 E0/1 and SW1 VLAN 20 SVI, notice that R5 takes a much longer path to reach the destination:

R5 -> R3 -> R1 -> SW1

Part 2: Tune OSPFv3

In this part of the lab, you will tune OSPF parameters, such as cost, interface priority, and passive interfaces.

Step 1

Configure the SW1 VLAN 20 SVI as a passive interface. By default, a router with OSPFv3 enabled on an interface begins sending hellos to the all-OSPF-routers multicast address (FF02::5) to establish a neighbor adjacency with any other OSPF-enabled router on the segment with matching hello parameters. SW1's VLAN 20 SVI should advertise the prefix (and it does) but it should not send hellos in VLAN 20 since there are no OSPF-enabled routers present.

```
SW1# config t
Enter configuration commands, one per line.  End with CNTL/Z.
SW1(config)# ipv6 router ospf 1
SW1(config-rtr)# passive-interface vlan 20
```

SW1 will continue to advertise the VLAN 20 prefix (`2001:db8:acad:20::/64`) but will no longer send hellos in VLAN 20. Use the `show ipv6 ospf interface vlan 20` command to confirm the passive state of the interface.

```
SW1# show ipv6 ospf interface vlan 20
Vlan20 is up, line protocol is up
```

```
  Link Local Address FE80::4, Interface ID 8 (ios-if-index)
  Area 0, Process ID 1, Instance ID 0, Router ID 4.4.4.4
  Network Type BROADCAST, Cost: 1
  Transmit Delay is 1 sec, State DR, Priority 1
  Designated Router (ID) 4.4.4.4, local address FE80::4
  No backup designated router on this network
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    No Hellos (Passive interface)
  Graceful restart helper support enabled
  Index 1/2/2, flood queue length 0
  Next 0x0(0)/0x0(0)/0x0(0)
  Last flood scan length is 0, maximum is 0
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 0, Adjacent neighbor count is 0
  Suppress hello for 0 neighbor(s)
```

Step 2

Configure a higher cost on the R1-R2 link so that R1 prefers the R1-R3 link when it wants to reach the `2001:db8:acad:23::/64` prefix. Recall in Part 1 when we discussed the OSPF cost. Currently, R1 load-balances through R2 and R3 to reach that prefix.

Use the `show ipv6 route 2001:db8:acad:23::/64` command to confirm that both paths currently have a metric of 20.

> Note: You can enable the Packet Capture feature on the link between SW1 and R5 before and after enabling the `passive interface` command and filter the output using the `proto ospf` BPF filter. The output from the packet capture should show you that OSPF Hello packets are no longer being advertised into VLAN20.

```
R1# show ipv6 route 2001:db8:acad:23::/64
Routing entry for 2001:DB8:ACAD:23::/64
  Known via "ospf 1", distance 110, metric 20, type intra area
  Route count is 2/2, share count 0
  Routing paths:
    FE80::3, Ethernet0/1
      Route metric is 20, traffic share count is 1
      From FE80::3
      Last updated 23:58:17 ago
    FE80::2, Ethernet0/0
      Route metric is 20, traffic share count is 1
      From FE80::2
      Last updated 23:58:17 ago
```

Use the interface configuration command `ipv6 ospf cost 500` to increase the cost on the R1-R2 link.

```
R1(config)# interface e0/0
R1(config-if)# ipv6 ospf cost 500

R2(config)# interface e0/0
R2(config-if)# ipv6 ospf cost 500
```

Use the `show ipv6 route 2001:db8:acad:23::/64` command again to confirm that only one path through R3 is now being used to reach the prefix with a cost of 20.

```
R1# show ipv6 route 2001:db8:acad:23::/64
Routing entry for 2001:DB8:ACAD:23::/64
  Known via "ospf 1", distance 110, metric 20, type intra area
  Route count is 1/1, share count 0
  Routing paths:
    FE80::3, Ethernet0/1
      Route metric is 20, traffic share count is 1
      From FE80::3
      Last updated 1d00h ago
```

If ever the link between R1 and R3 goes down, R1 will use the less-prefered link through R2 with a total cost of 510 to reach that prefix.

Step 3

Configure the SW1 VLAN 100 SVI so that it never becomes DR or BDR on the link with R1. Start by verifying the current state of the interface with the `show ipv6 ospf interface vlan 100`

```
SW1# show ipv6 ospf interface vlan 100
Vlan100 is up, line protocol is up
  Link Local Address FE80::4, Interface ID 9 (ios-if-index)
  Area 0, Process ID 1, Instance ID 0, Router ID 4.4.4.4
  Network Type BROADCAST, Cost: 1
  Transmit Delay is 1 sec, State DR, Priority 1
  Designated Router (ID) 4.4.4.4, local address FE80::4
  Backup Designated router (ID) 1.1.1.1, local address FE80::1
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:04
  Graceful restart helper support enabled
  Index 1/3/3, flood queue length 0
  Next 0x0(0)/0x0(0)/0x0(0)
  Last flood scan length is 0, maximum is 5
  Last flood scan time is 0 msec, maximum is 1 msec
  Neighbor Count is 1, Adjacent neighbor count is 1
    Adjacent with neighbor 1.1.1.1  (Backup Designated Router)
  Suppress hello for 0 neighbor(s)
```

The output above confirms that SW1 (4.4.4.4) is the DR while R1 (1.1.1.1) is the BDR. As both devices are currently configured with a default OSPF priority of 1, the router ID was used as a tie breaker, with higher values being preferred. Your results may vary.

Use the interface-level `ipv6 ospf priority 0` command on the SW1 VLAN 100 SVI to disable its particiaption in the DR/BDR election process. This will allow R1 to assume the role of the DR for the link, with no BDR being elected.

```
SW1(config)# interface vlan 100
SW1(config-if)# ipv6 ospf priority 0
```

Verify the results by using the same `show ipv6 ospf interface vlan 100` command on SW1.

```
SW1# show ipv6 ospf int vlan 100
Vlan100 is up, line protocol is up
  Link Local Address FE80::4, Interface ID 9 (ios-if-index)
  Area 0, Process ID 1, Instance ID 0, Router ID 4.4.4.4
  Network Type BROADCAST, Cost: 1
  Transmit Delay is 1 sec, State DROTHER, Priority 0
  Designated Router (ID) 1.1.1.1, local address FE80::1
  No backup designated router on this network
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:03
  Graceful restart helper support enabled
  Index 1/3/3, flood queue length 0
  Next 0x0(0)/0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 5
  Last flood scan time is 0 msec, maximum is 1 msec
  Neighbor Count is 1, Adjacent neighbor count is 1
    Adjacent with neighbor 1.1.1.1  (Designated Router)
  Suppress hello for 0 neighbor(s)
```

SW1 takes on a state called DROTHER (neither DR or BDR), R1 is the new DR, and there is no BDR on the link.

Part 3: Observe legitimate IPv6 SLAAC and Router Advertisement (RA)

In this part of the lab, you will enable the R2 Ethernet 0/2 interface and observe the IPv6 SLAAC and RA process that occurs between SW1 and R2.

Step 1

First, enable `debug ipv6 nd` on SW1 to analyze the legitimate IPv6 Neighbor Discovery messages that will be exchanged between SW1 and R2 in VLAN 20, and then enable R2's Ethernet 0/2 interface. Disable debugging after observing the ND and RA exhange between SW1 and R2.

```
SW1# debug ipv6 nd
  ICMP Neighbor Discovery events debugging is on
  ICMP ND HA events debugging is ON
SW1#

R2(config)# interface ethernet0/2
R2(config-if)# no shutdown
R2(config-if)#
*Sep 12 13:05:12.162: %LINK-3-UPDOWN: Interface Ethernet0/2, changed state to up
R2(config-if)#
*Sep 12 13:05:13.162: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/2, changed state to up


SW1#
*Sep 12 13:05:26.299: ICMPv6-ND: (Vlan20,FE80::2) Received NA from FE80::2
*Sep 12 13:05:26.299: ICMPv6-ND: Validating ND packet options: valid
*Sep 12 13:05:26.693: ICMPv6-ND: (Vlan20,FE80::2) Received RS
*Sep 12 13:05:26.693: ICMPv6-ND: Validating ND packet options: valid
*Sep 12 13:05:26.693: ICMPv6-ND: (Vlan20,FE80::2) Glean
*Sep 12 13:05:26.693: ICMPv6-ND: (Vlan20,FE80::2) LLA aabb.cc00.0520
*Sep 12 13:05:26.693: ICMPv6-ND: (Vlan20,FE80::2) INCMP -> STALE
*Sep 12 13:05:26.693: ICMPv6-ND: (Vlan20) Sending solicited RA
*Sep 12 13:05:26.693: ICMPv6-ND: (Vlan20) The interface media type supports ND resolution.
SW1#
*Sep 12 13:05:26.693: ICMPv6-ND: (Vlan20,FE80::4) send RA to FF02::1
*Sep 12 13:05:26.693: ICMPv6-ND: (Vlan20,FE80::4) Sending RA (1800) to FF02::1
*Sep 12 13:05:26.693: ICMPv6-ND:   MTU = 1500
*Sep 12 13:05:26.693: ICMPv6-ND:   prefix 2001:DB8:ACAD:20::/64 [LA] 2592000/604800
*Sep 12 13:05:27.696: ICMPv6-ND: (Vlan20,2001:DB8:ACAD:20::2) Received NA from 2001:DB8:ACAD:20::2
*Sep 12 13:05:27.696: ICMPv6-ND: Validating ND packet options: valid
SW1# undebug all
```

The highlighted portion shows SW1 receiving a router solicitation message from R2 and replying with an RA that carries the VLAN 20 prefix. A few moments later, SW1 receives a neighbor advertisement (NA) message from R2 confirming the SLAAC address R2 has assigned itself (`2001:db8:acad:20::2`).

Step 2

Return to R2 and verify the VLAN 20 prefix information that was applied to the E0/2 interface and verify the IPv6 routing table.

```
R2# show ipv6 interface ethernet0/2
Ethernet0/2 is up, line protocol is up
  IPv6 is enabled, link-local address is FE80::2
  No Virtual link-local address(es):
  Description: Link to SW1 Et0/1 - VLAN 20 SLAAC HOST simulation
  Stateless address autoconfig enabled
  Global unicast address(es):
    2001:DB8:ACAD:20::2, subnet is 2001:DB8:ACAD:20::/64 [EUI/CAL/PRE]
      valid lifetime 2591963 preferred lifetime 604763
  Joined group address(es):
    FF02::1
    FF02::2
    FF02::1:FF00:2
  MTU is 1500 bytes
  ICMP error messages limited to one every 100 milliseconds
  ICMP redirects are enabled
  ICMP unreachables are sent
  ND DAD is enabled, number of DAD attempts: 1
```

```
  ND reachable time is 30000 milliseconds (using 30000)
  ND RAs are suppressed (all)
  Hosts use stateless autoconfig for addresses.
```

Notice that R2's E0/2 interace has assigned itself the `2001:db8:acad:20::2/64` global unicast address through the stateless address autoconfiguration (SLAAC) process.

```
R2# show ipv6 route nd
IPv6 Routing Table - default - 16 entries
Codes: C - Connected, L - Local, S - Static, U - Per-user Static route
       B - BGP, R - RIP, H - NHRP, HG - NHRP registered
       Hg - NHRP registration summary, HE - NHRP External, I1 - ISIS L1
       I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary, D - EIGRP
       EX - EIGRP external, ND - ND Default, NDp - ND Prefix, DCE - Destination
       NDr - Redirect, RL - RPL, O - OSPF Intra, OI - OSPF Inter
       OE1 - OSPF ext 1, OE2 - OSPF ext 2, ON1 - OSPF NSSA ext 1
       ON2 - OSPF NSSA ext 2, la - LISP alt, lr - LISP site-registrations
       ld - LISP dyn-eid, lA - LISP away, le - LISP extranet-policy
       lp - LISP publications, ls - LISP destinations-summary, a - Application
       m - OMP
ND  ::/0 [2/0]
     via FE80::4, Ethernet0/2
NDp 2001:DB8:ACAD:20::/64 [2/0]
     via Ethernet0/2, directly connected
```

Notice that there are two entries in the IPv6 routing table related to the Neighbor Discovery (ND) process. Both entries are based on information carried in the RAs sent from SW1. The first entry is the default route (default gateway) pointing to SW1's VLAN 20 link-local address ``FE80::4``. The second is the global unicast prefix for the VLAN (``2001:db8:acad:20::/64``) ### Step 3 ### One final verification step before launching the attack in the Part 4 is to confirm the SW1 IPv6 preference level for RAs received by R2:

```
R2# show ipv6 routers ethernet0/2
Router FE80::4 on Ethernet0/2, last update 0 min
  Hops 64, Lifetime 1800 sec, AddrFlag=0, OtherFlag=0, MTU=1500
  HomeAgentFlag=0, Preference=Medium
  Reachable time 0 (unspecified), Retransmit time 0 (unspecified)
  Prefix 2001:DB8:ACAD:20::/64 onlink autoconfig
    Valid lifetime 2592000, preferred lifetime 604800
```

IPv6 Router Advertisement (RA) preference levels (defined in RFC 4191) allow routers to signal their priority as default gateways to host devices on the same link. There are three preference levels:

- High: Devices prefer this router first when building their default gateway list
- Medium: The default setting; hosts use this if no High-preference router exists. This is also the default value on Cisco routers an switches.
- Low: Used as a last resort option only when higher-preference routers are missing

Part 4

In this part of the lab, you will connect the rogue RA device to VLAN 20 and configure it to advertise higher priority RAs that will cause R5 E0/1 to offer itself as a better default gateway for the VLAN 20 network.

Step 1

Start by enabling `debug ipv6 nd` on R2. Then on R5 Ethernet0/1, adjust the RA interval to 10 seconds down from the default value of 200 seconds. This ensures that the attack will occur quickly when the interface is enabled. Increase the router preference to `high` to ensure that R5 becomes the new preferred default gateway for VLAN 20. FInally, enable the interface.

```
R2# debug ipv6 nd

R5(config)# interface ethernet0/1
R5(config-if)# ipv6 nd ra interval 10
R5(config-if)# ipv6 nd router-preference high
R5(config-if)# no shutdown
```

Return to the R2 console and observe the `debug` output. After a few moments, once the debug messages stop, turn off all debugging on R2. You should see following results:

```
R2#
*Sep 12 17:24:33.812: ICMPv6-ND: (Ethernet0/2,FE80::5) Received NA from FE80::5
*Sep 12 17:24:33.812: ICMPv6-ND: Validating ND packet options: valid
*Sep 12 17:24:33.812: ICMPv6-ND: (Ethernet0/2,FE80::5) Received RA
*Sep 12 17:24:33.812: ICMPv6-ND: Validating ND packet options: valid
*Sep 12 17:24:33.812: ICMPv6-ND: (Ethernet0/2,FE80::5) Glean
*Sep 12 17:24:33.812: ICMPv6-ND: (Ethernet0/2,FE80::5) LLA aabb.cc00.0910
*Sep 12 17:24:33.812: ICMPv6-ND: (Ethernet0/2,FE80::5) INCMP -> STALE
*Sep 12 17:24:33.812: ICMPv6-ND: [default] New router interface context created/74111D29C5D8
*Sep 12 17:24:33.812: ICMPv6-ND: [default] inserted router FE80::5/Ethernet0/2
*Sep 12 17:24:33.812: ICMPv6-ND: [default] Select default router
*Sep 12 17:24:33.812: ICMPv6-ND: [default] best rank is A19
*Sep 12 17:24:33.812: ICMPv6-ND: [default] router FE80::4/Ethernet0/2 no longer best
*Sep 12 17:24:33.812: ICMPv6-ND: [default] router FE80::5/Ethernet0/2 is new best
*Sep 12 17:24:33.812: ICMPv6-ND: [default] Selected new default router
*Sep 12 17:24:33.812: ICMPv6-ND: [default] Install default to FE80::5/Ethernet0/2
*Sep 12 17:24:33.812: ICMPv6-ND: Prefix : 2001:DB8:BAD:20::, Length: 64, Vld Lifetime: 2592000, Prf Lifetime: 604800, PI Flags: C0
R2#
*Sep 12 17:24:33.812: ICMPv6-ND: New on-link prefix 2001:DB8:BAD:20::/64 on Ethernet0/2/FE80::5, lifetime 2592000
*Sep 12 17:24:33.812: IPv6-Addrmgr-ND: DAD request for 2001:DB8:BAD:20::2 on Ethernet0/2
*Sep 12 17:24:33.812: ICMPv6-ND: (Ethernet0/2,2001:DB8:BAD:20::2) Sending DAD NS [Nonce: d32a.f646.a3d0]
*Sep 12 17:24:33.812: ICMPv6-ND: Autoconfiguring 2001:DB8:BAD:20::2 on Ethernet0/2
*Sep 12 17:24:34.811: ICMPv6-ND: (Ethernet0/2,2001:DB8:BAD:20::1) Received NA from 2001:DB8:BAD:20::1
*Sep 12 17:24:34.811: ICMPv6-ND: Validating ND packet options: valid
*Sep 12 17:24:34.812: IPv6-Addrmgr-ND: DAD: 2001:DB8:BAD:20::2 is unique.
R2#
*Sep 12 17:24:34.812: ICMPv6-ND: (Ethernet0/2,2001:DB8:BAD:20::2) Sending NA to FF02::1
```

```
R2#
*Sep 12 17:24:36.928: ICMPv6-ND: (Ethernet0/2,FE80::5) Received RA
*Sep 12 17:24:36.928: ICMPv6-ND: Validating ND packet options: valid
*Sep 12 17:24:36.928: ICMPv6-ND: Prefix : 2001:DB8:BAD:20::, Length: 64, Vld Lifetime: 2592000, Prf Lifetime: 604800, PI Flags: C0
*Sep 12 17:24:36.928: ICMPv6-ND: Update on-link prefix 2001:DB8:BAD:20::/64 on Ethernet0/2/FE80::5, lifetime 2592000
*Sep 12 17:24:36.928: ICMPv6-ND: %Ethernet0/2: OK: IPv6 Address Autoconfig 2001:DB8:BAD:20::/64 eui-64, 2001:DB8:BAD:20::2 2001:DB8:BAD:20::2/64 is existir
R2#
R2# undebug all
```

Let's analyze the highlighted output:

1. The first highlighted line indicates that R2 has received an RA from R5 on its E0/2 interface.
2. The second and third highlighted lines show that SW1 is no longer the best router. It has been replaced by the R5 router.
3. The fourth highlighted line confirms that R2 will now use R5 as its new preferred default gateway.
4. The fifth highlighted line indicates that R2 has learned a new *bad* prefix on E0/2 of `2001:db8:bad:20::/64`.
5. Finally, the last highlighted line shows R2 E0/2 using SLAAC to *autoconfigure* itself an IPv6 address of `2001:db8:bad:20::2` on the new *bad* prefix.

Verify the IPv6 parameters of R2's Ethernet0/2 interface, R2's IPv6 routing table and the state of the IPv6 routers that were learned on Ethernet0/2 :

```
R2# show ipv6 interface ethernet0/2
Ethernet0/2 is up, line protocol is up
  IPv6 is enabled, link-local address is FE80::2
  No Virtual link-local address(es):
  Description: Link to SW1 Et0/1 - VLAN 20 SLAAC HOST simulation
  Stateless address autoconfig enabled
  Global unicast address(es):
    2001:DB8:BAD:20::2, subnet is 2001:DB8:BAD:20::/64 [EUI/CAL/PRE]
      valid lifetime 2591993 preferred lifetime 604793
    2001:DB8:ACAD:20::2, subnet is 2001:DB8:ACAD:20::/64 [EUI/CAL/PRE]
      valid lifetime 2591851 preferred lifetime 604651
  Joined group address(es):
    FF02::1
    FF02::2
    FF02::1:FF00:2
  MTU is 1500 bytes
  ICMP error messages limited to one every 100 milliseconds
  ICMP redirects are enabled
  ICMP unreachables are sent
  ND DAD is enabled, number of DAD attempts: 1
  ND reachable time is 30000 milliseconds (using 30000)
  ND RAs are suppressed (all)
  Hosts use stateless autoconfig for addresses.
```

Notice that the R2 E0/2 interface now has two global unicast addresses: `2001:db8:bad:20::2/64` and `2001:db8:acad:20::2/64`. Both prefixes were learned via SLAAC.

```
R2# show ipv6 route nd
IPv6 Routing Table - default - 18 entries

<... output omitted ...>

ND  ::/0 [2/0]
     via FE80::5, Ethernet0/2
NDp 2001:DB8:BAD:20::/64 [2/0]
     via Ethernet0/2, directly connected
NDp 2001:DB8:ACAD:20::/64 [2/0]
     via Ethernet0/2, directly connected
```

Notice that the default route now points to R5 (`FE80::5`) and that the new *bad* prefix is directly connected to Ethernet0/2 along with the original prefix.

```
R2# show ipv6 routers ethernet0/2
Router FE80::5 on Ethernet0/2, last update 0 min
  Hops 64, Lifetime 1800 sec, AddrFlag=0, OtherFlag=0, MTU=1500
  HomeAgentFlag=0, Preference=High
  Reachable time 0 (unspecified), Retransmit time 0 (unspecified)
  Prefix 2001:DB8:BAD:20::/64 onlink autoconfig
    Valid lifetime 2592000, preferred lifetime 604800
Router FE80::4 on Ethernet0/2, last update 1 min
  Hops 64, Lifetime 1800 sec, AddrFlag=0, OtherFlag=0, MTU=1500
  HomeAgentFlag=0, Preference=Medium
  Reachable time 0 (unspecified), Retransmit time 0 (unspecified)
  Prefix 2001:DB8:ACAD:20::/64 onlink autoconfig
    Valid lifetime 2592000, preferred lifetime 604800
```

The output confirms that R5 is advertising a *bad* prefix with a higher preference than the legitimate `2001:db8:acad:20::/64` prefix from SW1.

Part 5

In the final part of the lab, you will implement an RA Guard policy on SW1 that will prevent R5 from advertising bogus RAs into VLAN 20.

RA Guard is a switch feature that inspects RAs arriving on a port and drops them if that port isn't supposed to have a router behind it. RA Guard defines two switch port roles:

- **router**
- **host**

The **router** role is used for ports where a legitimate router is connected (usually an uplink) and RAs are permitted to arrive on that port. The **host** role is used for ports where hosts are connected (PCs, printers, phones) and RAs are dropped if they arrive on those ports.

The switch examines incoming ICMPv6 traffic at Layer 2. When it sees an RA on a **host** port, it discards the packet.

Configuring RA Guard is a two-step process. First, policies are defined for both **router** and **host** roles, and then those policies are applied either globally to a VLAN or specifically to a physical port.

Step 1

Start by defining the **router** and **host** policies on SW1.

```
SW1(config)# ipv6 nd raguard policy HOST-POLICY
SW1(config-ra-guard)#  device-role host
SW1(config-ra-guard)#  exit
SW1(config)# ipv6 nd raguard policy ROUTER-POLICY
SW1(config-ra-guard)#  device-role router
SW1(config-ra-guard)#  exit
```

Step 2

Next, apply the **host** policy globally to the entire VLAN 20.

```
SW1(config)# vlan configuration 20
SW1(config-vlan-config)#  ipv6 nd raguard attach-policy HOST-POLICY
SW1(config-vlan-config)# exit
```

Although it is possible to apply the **host** policy to individual interfaces, applying it globally to the entire VLAN is a recommended best practice.

Step 3

Since SW1 is the trusted default gateway, legitimate RAs originate from SW1's VLAN 20 SVI and egress the access ports. RA Guard never inspects the RAs from SW1, so the **router** policy isn't required or necessary for VLAN 20. As an example though, the **router** policy can be enabled on the Ethernet0/0 interface in VLAN 100 so as to trust the RAs from R1.

```
SW1(config)# interface Ethernet0/0
SW1(config-if)# ipv6 nd raguard attach-policy ROUTER-POLICY
```

You can use the `show ipv6 nd raguard policy` command to validate the RA Guard configuration.

```
SW1# show ipv6 nd raguard policy
RA guard policy HOST-POLICY configuration:
  device-role host
Policy HOST-POLICY is applied on the following targets:
Target               Type  Policy               Feature        Target range
vlan 20              VLAN  HOST-POLICY          RA guard       vlan all

RA guard policy ROUTER-POLICY configuration:
  device-role router
Policy ROUTER-POLICY is applied on the following targets:
Target               Type  Policy               Feature        Target range
Et0/0                PORT  ROUTER-POLICY        RA guard       vlan all
```

Step 4

With RA Guard implemented, test the solution. Start by enabling `debug ipv6 snooping raguard` on SW1. Within a few seconds you should see that RA Guard has already started blocking bogus RAs from R5:

```
SW1# debug ipv6 snooping raguard
  Device-tracking - RA guard debugging is on
SW1#
*Sep 12 18:51:18.269: SISF[RAG]: Et0/2 vlan 20 RA received by RA guard on Et0/2 from FE80::5
*Sep 12 18:51:18.269: SISF[RAG]: Et0/2 vlan 20     option 1 : ND_OPT_SOURCE_LINKADDR
*Sep 12 18:51:18.269: SISF[RAG]: Et0/2 vlan 20     option 3 : ND_OPT_PREFIX_INFORMATION
*Sep 12 18:51:18.269: SISF[RAG]: Et0/2 vlan 20     option 5 : ND_OPT_MTU
*Sep 12 18:51:18.269: SISF[RAG]: Et0/2 vlan 20 !Not a router port: all router messages disallowed
*Sep 12 18:51:18.269: SISF[RAG]: Et0/2 vlan 20 ! DROP ROUTER-ADVERT  src FE80::5 dst FF02::1 reason = 3
SW1#
```

The output confirms that Ethernet0/2 in VLAN 20 is not a **router** port and that all RAs from R5 are dropped.

If you leave the `debug` command enabled for longer, you might see that RAs received on VLAN 100 from R1 and RAs sourced from SW1 VLAN 20 SVI are allowed, as expected.

Step 5

With the bogus RAs from R5 being filtered on SW1, reset the R2 Ethernet0/2 interface to clear the *bad* prefix and default route, and then verify the results.

```
R2(config)# interface ethernet0/2
R2(config-if)# shutdown
R2(config-if)#
*Sep 12 18:38:46.917: %LINK-5-CHANGED: Interface Ethernet0/2, changed state to administratively down
*Sep 12 18:38:47.917: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/2, changed state to down
R2(config-if) # no shutdown
R2(config-if)#
*Sep 12 18:39:10.295: %LINK-3-UPDOWN: Interface Ethernet0/2, changed state to up
R2(config-if)#
*Sep 12 18:39:11.295: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/2, changed state to up
R2(config-if)# end
R2#
R2# show ipv6 routers ethernet0/2
Router FE80::4 on Ethernet0/2, last update 0 min
  Hops 64, Lifetime 1800 sec, AddrFlag=0, OtherFlag=0, MTU=1500
  HomeAgentFlag=0, Preference=Medium
```

```
    Reachable time 0 (unspecified), Retransmit time 0 (unspecified)
    Prefix 2001:DB8:ACAD:20::/64 onlink autoconfig
      Valid lifetime 2592000, preferred lifetime 604800
R2#
R2# show ipv6 int ethernet0/2
Ethernet0/2 is up, line protocol is up
  IPv6 is enabled, link-local address is FE80::2
  No Virtual link-local address(es):
  Description: Link to SW1 Et0/1 - VLAN 20 SLAAC HOST simulation
  Stateless address autoconfig enabled
  Global unicast address(es):
    2001:DB8:ACAD:20::2, subnet is 2001:DB8:ACAD:20::/64 [EUI/CAL/PRE]
      valid lifetime 2591929 preferred lifetime 604729
  Joined group address(es):
    FF02::1
    FF02::2
    FF02::1:FF00:2
  MTU is 1500 bytes
  ICMP error messages limited to one every 100 milliseconds
  ICMP redirects are enabled
  ICMP unreachables are sent
  ND DAD is enabled, number of DAD attempts: 1
  ND reachable time is 30000 milliseconds (using 30000)
  ND RAs are suppressed (all)
  Hosts use stateless autoconfig for addresses.

R2# show ipv6 route nd
IPv6 Routing Table - default - 16 entries

<... output omitted ...>

ND  ::/0 [2/0]
     via FE80::4, Ethernet0/2
NDp 2001:DB8:ACAD:20::/64 [2/0]
     via Ethernet0/2, directly connected
```

R2's Ethernet 0/2 is now reporting only one IPv6 global unicast prefix (``2001:db8:acad:20::/64) being learned from SW1 through SLAAC and only one global unicast address in that prefix. The IPv6 routing table now shows only the appropriate ND-learned default route pointing to SW1.

The End

Congratulations! You have completed the lab! You learned how to configure, tune and verify OSPFv3, as well as how to deploy RA Guard.